

Notes on the Construction of Tangent Spaces of Character Varieties

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1 Introduction

In addition to notes for my talk, these notes can also be thought of as a companion to Chapters 3 and 4 of [3].

Definition 1.1. Let Γ be a discrete group and let G be a linear Lie group. The *representation variety* of Γ into G is the space $\text{Rep}(\Gamma, G) := \text{Hom}(\Gamma, G)/G$, where G acts on $\text{Hom}(\Gamma, G)$ by conjugation. Recall that the *character* χ_ρ of a representation $\rho : \Gamma \rightarrow G$ of Γ is the class function $\chi_\rho(\gamma) := \text{tr}(\rho(\gamma))$. Let $\sim$ be the equivalence relation on $\text{Hom}(\Gamma, G)$ where $\rho_1 \sim \rho_2$ when $\chi_{\rho_1} = \chi_{\rho_2}$. The *character variety* of Γ into G is the space $\chi(\Gamma, G) := \text{Rep}(\Gamma, G)/\sim$.

Notice that we have not defined a topology on $\text{Rep}(\Gamma, G)$ or on $\chi(\Gamma, G)$. Instead, we will use the following guiding principle: whatever the topology on $\chi(\Gamma, G)$ may be, there should be a tangent space $T_{[\rho]}\chi(\Gamma, G)$ for every equivalence class $[\rho] \in \chi(\Gamma, G)$ whose elements can be described explicitly as “infinitesimal directions in which ρ may vary.” The goal of these notes is to describe rigorously the construction of a suitable set of “infinitesimal directions” in the case where Γ is the fundamental group $\pi_1(S)$ of a surface S . We will see that this set is some sort of cohomology group, so that we have:

Theorem 1.2. *Given a surface S , a linear Lie group G with Lie algebra $\mathfrak{g}$, and a representation $\rho : \pi_1(S) \rightarrow G$, there is a natural isomorphism*

$$T_{[\rho]}\chi(\pi_1(S), G) \cong H_\rho^1(\mathfrak{g}).$$

It is therefore the goal of these notes to define the “twisted” cohomology group $H_\rho^1(\mathfrak{g})$ (Definition 3.18).

2 The de Rham Version of Twisted Cohomology

There are several ways to go about defining $H^1_\rho(\mathfrak{g})$. One way that showcases the geometric nature of this theory is the de Rham-style interpretation. The key ingredient here is the notion of holonomy. Recall that in Riemannian geometry, a Riemannian metric on a manifold M induces a connection that allows us to lift paths (“parallel transport”) from M to TM in a way uniquely determined by an initial lift. The parallel transport depends only on the connection, not on the metric, and so we should expect that a connection is the sort of thing that allows us to uniquely lift paths along any bundle projection given an initial lift. Indeed this is the case, and we introduce the construction here. Let us first take a minute to review some basic concepts and fix the notation that will be used in these notes.

Given a smooth map $f : M \rightarrow N$ of manifolds, we will write the derivative (i.e. pushforward) of f at $p \in M$ as $f_* : T_p M \rightarrow T_{f(p)} N$. Often we will write $(f_*)_p$ instead to keep track of the point p . We will write the exterior derivative of f as $df \in T^* M$, and we will write the value of df at a point $p \in M$ as $d_p f \in T_p^* M$. All vector spaces we consider will be over $\mathbb{R}$.

Given a fiber bundle $\xi : F \xrightarrow{\pi} M$, where F is the total space of ξ , we will simply refer to ξ as $F \rightarrow M$ or F . For example, given vector bundles $E \rightarrow M$ and $F \rightarrow M$, we will write $E \otimes F$ for the tensor product of the two vector bundles. We recall the following definition.

Definition 2.1. Given a fiber bundle $E \xrightarrow{\pi} M$ and a smooth map $f : N \rightarrow M$ of manifolds, the *pullback bundle* $f^* E$ is the set $\{(n, e) \in N \times E \mid f(n) = \pi(e)\}$. We topologize $f^* E$ as a submanifold of $N \times E$, and there are canonical maps $f^* E \ni (n, e) \mapsto n \in N$ and $f^* E \ni (n, e) \mapsto e \in E$ such that the diagram

$$\begin{array}{ccc} f^* E & \longrightarrow & E \\ \downarrow & & \downarrow \pi \\ N & \xrightarrow{f} & M \end{array}$$

commutes. Observe that given $O \xrightarrow{g} N \xrightarrow{f} M$, we have $g^* f^* E = (f \circ g)^* E$. Also observe that if E is a vector bundle, then so is $f^* E$.

2.1 Connections and Holonomy

In what follows, let $E \xrightarrow{\pi} M$ be some vector bundle over a manifold M . This subsection is basically just a brief paraphrase of the parts of Chapter 3 of [3] that don’t discuss structure groups. We will diverge from [3] in our discussion of structure groups.

Definition 2.2. A section of $E \otimes \bigwedge^n T^* M$ is called a *differential E -valued n -form*, and the set $\Gamma(E \otimes \bigwedge^n T^* M)$ of such sections is denoted by $\Omega^n(M, E)$. We also denote by $\Omega^*(M, E) := \bigoplus_{i=1}^\infty \Omega^i(M, E)$ the graded $C^\infty(M)$ -module of differential E -valued forms.

The last part of the above definition makes sense, since for any vector bundle $F \rightarrow M$, the set $\Gamma(F)$ of sections of F has a natural $C^\infty(M)$ -module structure given by fiberwise addition and scalar multiplication.

Remark 2.3. Notice that real-valued differential n -forms are a particular case of the above construction: they can be identified with differential $(M \times \mathbb{R})$ -valued n -forms, where $M \times \mathbb{R}$ is the trivial line bundle over M . We will write $\Omega^n(M)$ for $\Omega^n(M, M \times \mathbb{R})$, the $C^\infty(M)$ -module of real-valued differential n -forms. Finally, notice that $\Omega^0(M, E)$ is simply $\Gamma(E)$, the set of sections of E .

Definition 2.4. A *connection*, or *covariant derivative*, on E is an $\mathbb{R}$ -linear map $\nabla : \Omega^0(M, E) \rightarrow \Omega^1(M, E)$ satisfying the Leibniz rule:

$$\nabla(f\sigma) = df \otimes \sigma + f\nabla\sigma, \quad \forall f \in C^\infty(M), \sigma \in \Omega^0(M, E).$$

Recall that for vector spaces V, W , there is a natural identification of $V \otimes \bigwedge^n W^*$ with the space of alternating n -linear maps $W \times \cdots \times W \rightarrow V$. Via this identification, every element α_p of the fiber of $E \otimes T^*M$ over p can be viewed as a linear map $\alpha_p : T_p M \rightarrow E_p$, where E_p is the fiber of E over p . Therefore, given a vector field $X \in \Gamma(TM)$ and a section $\sigma \in \Omega^0(M, E)$, we define $\nabla_X \sigma \in \Omega^1(M, E)$ by

$$(\nabla_X \sigma)_p := (\nabla \sigma)_p(X_p), \quad \forall p \in M.$$

Example 2.5.

- (i) Given a trivial vector bundle $M \times \mathbb{R}^n$, a section $\sigma : M \rightarrow M \times \mathbb{R}^n$ is completely determined by its values in the second coordinate. Therefore let us write such a section as $\sigma : M \rightarrow \mathbb{R}^n$. Then the *trivial connection* D is defined by $(D_X \sigma)_p := (\sigma_*)_p X \in T_{\sigma(p)} \mathbb{R}^n \cong \mathbb{R}^n \cong (M \times \mathbb{R}^n)_p$ for every $p \in M$, $X \in T_p M$.
- (ii) If M is equipped with a Riemannian metric, then the induced Levi-Civita connection is indeed a connection on TM in the sense of Definition 2.4.

The following is a crucial technical lemma whose proof is a straightforward verification.

Lemma 2.6. *The additive group $\Omega^1(M, \text{End}(E))$ acts transitively and freely on the set of connections on E . That is, given any connection ∇ on E and any $\text{End}(E)$ -valued 1-form $A \in \Omega^1(M, \text{End}(E))$, every other connection on E can be written uniquely as $\nabla + A$, where*

$$(\nabla + A)_X \sigma = \nabla_X \sigma + A(X)\sigma, \quad \forall X \in \Gamma(TM), \sigma \in \Omega^0(M, E).$$

These connections are exactly what we need if we want a unique way to lift paths from M to E given an initial lift. In what follows, let ∇ be a connection on E . For a path $\gamma : [0, 1] \rightarrow M$, we will need to say when a section $\sigma : [0, 1] \rightarrow \gamma^* E$ is “parallel” to the connection ∇ . Therefore we will need a way of pulling back ∇ to a connection $\gamma^* \nabla$ on $\gamma^* E$.

Proposition 2.7. *Given any smooth map $f : N \rightarrow M$ of manifolds and a connection ∇ on the vector bundle $E \rightarrow M$, there is a unique connection $f^* \nabla$ on the pullback bundle $f^* E \rightarrow N$ satisfying*

$$(f^* \nabla)_Y (f^* \sigma) = f^* (\nabla_{f_* Y} \sigma), \quad \forall Y \in \Gamma(TN), \sigma \in \Omega^0(M, E).$$

Proof sketch. On charts U over which E is trivialized, by Lemma 2.6 there is some $A \in \Omega^1(U, \text{End}(E|_U))$ for which $\nabla = D + A$, where D is the trivial connection on $E|_U$. Then define $f^* \nabla$ on $f^{-1}(U)$ by $f^* \nabla = D + f^* A$, where D now denotes the trivial connection on $f^* E|_{f^{-1}(U)}$. One checks that this is the unique connection on $f^* E|_{f^{-1}(U)}$ satisfying the above condition, and so we have defined $f^* \nabla$ independently of the chosen chart and trivialization. $\square$

Definition 2.8. Given a smooth map $f : N \rightarrow M$ of manifolds and a connection ∇ on the vector bundle $E \rightarrow M$, a section $\sigma : N \rightarrow f^* E$ is *parallel* to ∇ if $(f^* \nabla) \sigma = 0 \in \Omega^1(N, f^* E)$.

In particular, for a path $\gamma : [0, 1] \rightarrow M$, a section $\sigma : [0, 1] \rightarrow \gamma^* E$ is parallel to ∇ if $(\gamma^* \nabla) \frac{\partial}{\partial t} \sigma \equiv 0$.

Proposition 2.9. *Given a path $\gamma : [0, 1] \rightarrow M$ and some $v \in E_{\gamma(0)} = (\gamma^* E)_0$, there is a unique section $\sigma_v : [0, 1] \rightarrow \gamma^* E$ parallel to ∇ such that $\sigma_v(0) = v$. Furthermore, $v \mapsto \sigma_v(1)$ is a linear isomorphism $E_{\gamma(0)} \rightarrow E_{\gamma(1)}$.*

Proving Proposition 2.9 is just a matter of invoking the existence and uniqueness theorem for ODEs, see [3] for details. Using the notation of Proposition 2.9, the composition $\tilde{\gamma} \circ \sigma_v$ is a lift of γ to E , where $\tilde{\gamma} : \gamma^* E \rightarrow E$ denotes the canonical map from the pullback $\gamma^* E$ to E .

Definition 2.10. Let $\gamma : [0, 1] \rightarrow M$ be a loop. Then the *holonomy isomorphism* determined by ∇ is the map

$$\text{Hol}^\nabla(\gamma) : E_{\gamma(0)} \rightarrow E_{\gamma(0)}, \quad \text{Hol}^\nabla(\gamma)(v) := \sigma_v(1).$$

2.2 The Covariant Exterior Derivative

We follow §19.13 of [4] closely in this section.

2.2.1 Definition and Basic Properties

Just as differential forms with values in vector bundles generalized ordinary differential forms, there is a “covariant” exterior derivative, depending on a connection ∇ on E , that generalizes the exterior derivative of differential forms. We will see that this covariant exterior derivative makes $\Omega^*(M, E)$ into a chain complex precisely when ∇ is a “flat” connection. Later, we will see that $H_\rho^1(\mathfrak{g})$ is a cohomology group that comes from such a chain complex.

This approach using differential forms (the “de Rham approach”) is only one of many equivalent ways of describing twisted cohomology. Other approaches are described in §4.2 of [3], and §§4.4-5 of [1]. The following definition uses an explicit formula for d_∇ . Note that in the case of the trivial connection $\nabla = D$, this is simply the coordinate-free formula for the ordinary exterior derivative of real-valued differential forms (see §9.8 of [4]). Note that the notation $\widehat{X}_i$ denotes the fact that X_i is *omitted* from the spot where it would otherwise be taken as an argument.

Definition 2.11. Let $E \rightarrow M$ be a vector bundle with a connection ∇ . The *covariant exterior derivative* $d_\nabla : \Omega^k(M, E) \rightarrow \Omega^{k+1}(M, E)$ is defined by the formula

$$\begin{aligned} d_\nabla \alpha(X_0, \dots, X_k) &:= \sum_{i=0}^k (-1)^i \nabla_{X_i} \alpha(X_0, \dots, \widehat{X}_i, \dots, X_k) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \alpha([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_k), \end{aligned}$$

for any $\alpha \in \Omega^k(M, E)$, $X_0, \dots, X_k \in \Gamma(TM)$. Notice that d_∇ is $\mathbb{R}$ -linear.

Observe that $\Omega^*(M, E) \cong \Omega^*(M) \otimes \Gamma(E)$ is a graded $\Omega^*(M)$ -module under the wedge product

$$\eta \wedge \left(\sum_{i=1}^n \omega_i \otimes \sigma_i \right) := \sum_{i=1}^n (\eta \wedge \omega_i) \otimes \sigma_i$$

for $\eta \in \Omega^*(M)$, $\omega_i \in \Omega^*(M)$, $\sigma_i \in \Gamma(E)$.

Proposition 2.12. Let $E \rightarrow M$ be a vector bundle with a connection ∇ . The covariant exterior derivative satisfies the following properties.

- (i) For $\sigma \in \Gamma(E) = \Omega^0(M, E)$, we have $d_\nabla \sigma = \nabla \sigma$.
- (ii) (Leibniz rule) For $\eta \in \Omega^k(M)$, $\alpha \in \Omega^\ell(M, E)$, we have

$$d_\nabla(\eta \wedge \alpha) = d\eta \wedge \alpha + (-1)^k \eta \wedge d_\nabla \alpha.$$

Proof. The formula in Definition 2.11 gives

$$d_\nabla \sigma(X) = \sum_{i=0}^0 (-1)^i \nabla_{X_i} \sigma + \sum \emptyset = \nabla_X \sigma$$

for every $X \in \Gamma(TM)$ ($\sum \emptyset = 0$ is the sum over the empty set). Therefore (i) holds.

We first make the following computation for simple tensors $\omega \otimes \sigma \in \Omega^k(M, E) \cong \Omega^k(M) \otimes \Gamma(E)$:

$$\begin{aligned}
d_{\nabla}(\omega \otimes \sigma) &= \sum_{i=0}^{\ell} (-1)^i \nabla_{X_i}(\omega \otimes \sigma)(X_0, \dots, \widehat{X}_i, \dots, X_{\ell}) \\
&\quad + \sum_{0 \leq i < j \leq \ell} (-1)^{i+j} (\omega \otimes \sigma)([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_{\ell}) \\
&= \sum_{i=1}^{\ell} (X_i(\omega(X_0, \dots, \widehat{X}_i, \dots, X_{\ell})))\sigma + \omega(X_0, \dots, \widehat{X}_i, \dots, X_{\ell})\nabla_{X_i}\sigma \\
&\quad + \sum_{0 \leq i < j \leq \ell} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_{\ell})\sigma \\
&= (d\omega \otimes \sigma)(X_0, \dots, X_{\ell}) + (-1)^{\ell} (\omega \wedge \nabla\sigma)(X_0, \dots, X_{\ell}).
\end{aligned}$$

Since $\nabla\sigma = d_{\nabla}\sigma$ by (i), the above expression is equal to $d_{\nabla}(\omega \otimes \sigma) = d\omega \otimes \sigma + (-1)^{\ell} \omega \wedge d_{\nabla}\sigma$. By the $\mathbb{R}$ -linearity of d_{∇} , it suffices to check (ii) for α a simple tensor $\omega \otimes \sigma$. We have

$$\begin{aligned}
d_{\nabla}(\eta \wedge (\omega \otimes \sigma)) &= d(\eta \wedge \omega) \otimes \sigma + (-1)^{k+\ell} (\eta \wedge \omega) \wedge d_{\nabla}\sigma \\
&= (d\eta \wedge \omega + (-1)^k \eta \wedge d\omega) \otimes \sigma + (-1)^{k+\ell} (\eta \wedge \omega) \wedge d_{\nabla}\sigma \\
&= d\eta \wedge (\omega \otimes \sigma) + (-1)^k \eta \wedge (d\omega \otimes \sigma + (-1)^{\ell} \omega \wedge d_{\nabla}\sigma) \\
&= d\eta \wedge (\omega \otimes \sigma) + (-1)^k \eta \wedge d_{\nabla}(\omega \otimes \sigma),
\end{aligned}$$

as desired. $\square$

Remark 2.13. The covariant exterior derivative d_{∇} is in fact the unique $\mathbb{R}$ -linear map $\Omega^k(M, E) \rightarrow \Omega^{k+1}(M, E)$ satisfying (i) and (ii) of Proposition 2.12.

2.2.2 The Curvature as an Obstruction

Definition 2.14. Let $E \rightarrow M$ be a vector bundle with a connection ∇ . Given $X, Y \in \Gamma(TM)$ and $\sigma \in \Gamma(E) = \Omega^0(M, E)$ we define

$$R^{\nabla}(X, Y)\sigma := \nabla_X \nabla_Y \sigma - \nabla_Y \nabla_X \sigma - \nabla_{[X, Y]}\sigma.$$

It is easily computed that R^{∇} is tensorial (i.e., $C^{\infty}(M)$ -linear) in X, Y , and σ . Hence R^{∇} can be understood as a differential $\text{End}(E)$ -valued 2-form $R^{\nabla} \in \Omega^2(M, \text{End}(E))$. There is an action of $\Omega^*(M, \text{End}(E))$ on $\Omega^*(M, E)$ given by, for $A \in \Omega^k(M, \text{End}(E)), \alpha \in \Omega^{\ell}(M, E)$,

$$(A \wedge \alpha)(X_1, \dots, X_{k+\ell}) := \frac{1}{k!\ell!} \sum_{s \in \text{Perm}(\{1, \dots, k+\ell\})} \text{sign}(s) (A(X_{s(1)}, \dots, X_{s(k)}) \circ \alpha)(X_{s(k+1)}, \dots, X_{s(k+\ell)}),$$

where $X_1, \dots, X_{k+\ell} \in \Gamma(TM)$, and $\text{Perm}(S)$ is the group of permutations of the finite set S .

The following proposition says that R^{∇} is an obstruction to

$$\dots \xrightarrow{d_{\nabla}} \Omega^{\ell-1}(M, E) \xrightarrow{d_{\nabla}} \Omega^{\ell}(M, E) \xrightarrow{d_{\nabla}} \Omega^{\ell+1}(M, E) \xrightarrow{d_{\nabla}} \dots$$

being a cochain complex.

Proposition 2.15. Let $E \rightarrow M$ be a vector bundle with a connection ∇ . For $\alpha \in \Omega^{\ell}(M, E)$, we have $d_{\nabla}^2 \alpha = R^{\nabla} \wedge \alpha$.

Proof. By the $\mathbb{R}$ -linearity of d_∇ , it suffices to check that the desired identity holds when α is a simple tensor $\alpha = \omega \otimes \sigma$. We have

$$\begin{aligned} d_\nabla^2(\omega \otimes \sigma) &= d_\nabla(d\omega \otimes \sigma + (-1)^\ell \wedge d_\nabla \sigma) \\ &= dd\omega \otimes \sigma + (-1)^{\ell+1} d\omega \wedge d_\nabla \sigma + (-1)^\ell d\omega \wedge d_\nabla \sigma + (-1)^{2\ell} \omega \wedge d_\nabla d_\nabla \sigma \\ &= \omega \wedge d_\nabla \nabla \sigma. \end{aligned}$$

For any $X_0, \dots, X_{\ell+1} \in \Gamma(TM)$, we therefore have

$$\begin{aligned} \omega \wedge d_\nabla \nabla \sigma(X_0, \dots, X_{\ell+1}) &= \frac{1}{\ell! \cdot 2} \sum_{s \in \text{Perm}(\{0, \dots, \ell+1\})} \text{sign}(s) \omega(X_{s(0)}, \dots, X_{s(\ell-1)}) (d_\nabla \nabla \sigma)(X_{s(\ell)}, X_{s(\ell+1)}) \\ &= \frac{1}{\ell! \cdot 2} \sum_{s \in \text{Perm}(\{0, \dots, \ell+1\})} \text{sign}(s) \omega(X_{s(0)}, \dots, X_{s(\ell-1)}) \left(\nabla_{X_{s(\ell)}} \nabla_{X_{s(\ell+1)}} \sigma - \nabla_{X_{s(\ell+1)}} \nabla_{X_{s(\ell)}} \sigma - \nabla_{[X_{s(\ell)}, X_{s(\ell+1)}]} \sigma \right) \\ &= \frac{1}{\ell! \cdot 2} \sum_{s \in \text{Perm}(\{0, \dots, \ell+1\})} \text{sign}(s) R^\nabla(X_{s(\ell)}, X_{s(\ell+1)}) (\omega(X_{s(0)}, \dots, X_{s(\ell-1)}) \sigma) \\ &= \frac{1}{\ell! \cdot 2} \sum_{s \in \text{Perm}(\{0, \dots, \ell+1\})} \text{sign}(s) (R^\nabla(X_{s(\ell)}, X_{s(\ell+1)}) \circ (\omega \otimes \sigma)) (X_{s(0)}, \dots, X_{s(\ell-1)}) \\ &= (R^\nabla \wedge (\omega \otimes \sigma)) (X_{s(0)}, \dots, X_{s(\ell+1)}). \end{aligned}$$

Therefore the desired identity holds when α is a simple tensor $\omega \otimes \sigma$, and hence holds for all $\alpha \in \Omega^\ell(M, E)$. $\square$

We say that a connection ∇ is *flat* if R^∇ is the zero section of the bundle $(\bigwedge^2 T^*M) \otimes \text{End}(E)$. It follows immediately from Proposition 2.15 that $\Omega^*(M, E)$ is a cochain complex with differential d_∇ when ∇ is flat.

Definition 2.16. Let $E \rightarrow M$ be a vector bundle with a connection ∇ . If ∇ is flat, then the *n th twisted cohomology group* $H_\nabla^n(M)$ is the n th cohomology group of the cochain complex $\Omega^*(M, E)$ with differential d_∇ .

The twisted cohomology group $H_\rho^1(\mathfrak{g})$ will be identified with $H_\nabla^1(S)$ for some suitable choice of vector bundle $E \rightarrow S$ and flat connection ∇ on E . It is the goal of the next section to describe these choices E and ∇ .

2.3 (+ ε) The Cartan Structure Equation

Suppose I write a connection locally as $\nabla = D + A$, where D is the trivial connection. How do I write down the curvature ∇^2 ? One can compute explicitly that for $\omega \in \Omega^k(M)$ and $e \in \Omega^\ell(M, E)$, we have $A \wedge (\omega \wedge e) = (-1)^k \omega \wedge (A \wedge e)$, and if $\ell = 1$ then $A \wedge (\omega \otimes e) = (-1)^k \omega \wedge (A \wedge e)$.

Lemma 2.17. *We have $d_\nabla = D + A \wedge \cdot$.*

Proof. It suffices to check the Leibniz rule:

$$\begin{aligned} (D + A \wedge \cdot)(\omega \otimes e) &= d\omega \otimes e + (-1)^k \omega \wedge De + (-1)^k \omega \wedge (A \wedge e) \\ &= d\omega \otimes e + (-1)^k \omega \wedge (D + A \wedge e). \end{aligned}$$

$\square$

Proposition 2.18 (Cartan Structure Equation).

$$\nabla^2 = DA + A \wedge A.$$

Proof. We have

$$\begin{aligned}
(D + A \wedge \cdot)(D + A \wedge \cdot)\alpha &= (D + A \wedge \cdot)(D\alpha + A \wedge \alpha) \\
&= D^2\alpha + D(A \wedge \alpha) + A \wedge D\alpha + A \wedge A \wedge \alpha \\
&= (DA)\alpha + A \wedge A \wedge \alpha.
\end{aligned}$$

□

3 Splittings of Bundles and Structure Groups

This section elaborates on the ideas laid out in §§8-9 in [5].

3.1 Connections as Splittings

Given a vector bundle $E \xrightarrow{\pi} M$ and a point $x \in E$, let $V_x := \ker(\pi_*)_x \subseteq T_x E$. Note that $V_x = T_x \pi^{-1}(\pi(x))$. Then $V := (\bigcup_{x \in E} V_x) \rightarrow E$ is a subbundle of $TE \rightarrow E$, called the *vertical subbundle*. Note that $(\pi^* TM)_x$ is simply a copy of $T_{\pi(x)} M$ sitting over $x \in E$, and so there is a short exact sequence

$$0 \rightarrow V_x \rightarrow T_x E \xrightarrow{(\pi_*)_x} (\pi^* TM)_x \rightarrow 0.$$

As x ranges over all of E , this gives a short exact sequence of vector bundles

$$0 \rightarrow V \rightarrow TE \rightarrow \pi^* TM \rightarrow 0.$$

Now let $p \in M$, let $x \in E_p = \pi^{-1}(p)$, and let ∇ be a connection on E . On a neighborhood U of p in M over which E is trivialized, we can find a section $\tilde{\sigma} \in \Gamma(E|_U)$ with $\tilde{\sigma}(p) = x$ and $\nabla \tilde{\sigma} = 0 \in \Omega^1(U, E|_U)$. Multiplying $\tilde{\sigma}$ by a bump function supported on U allows us to define a section $\sigma \in \Gamma(E)$ with $\sigma(p) = x$ and $\nabla \sigma = 0$ on a neighborhood of p . Note that $(\sigma_*)_p$ is a map $T_p M \rightarrow T_x E$, and that if $\tau \in \Gamma(E)$ is some other section with $\tau(p) = x$ and $\nabla \tau = 0$, then $\sigma - \tau = f\eta$ for some $\eta \in \Gamma(E)$ and some $f \in C^\infty(M)$ with $f(p) = 0$. Note that

$$0 = (\nabla(\sigma - \tau))_p = (\nabla(f\eta))_p = d_p f \otimes \eta(p) + f(p) \nabla \eta = \eta(p) d_p f.$$

Therefore

$$\begin{aligned}
(\sigma_*)_p X &= (\tau_* + (\sigma - \tau)_*)_p \\
&= (\tau_*)_p X + \eta(p) d_p f(X) + f(p) (\eta_*)_p(X) \\
&= (\tau_*)_p X
\end{aligned}$$

for every $X \in T_p M$. The following definition therefore makes sense.

Definition 3.1. Let $E \xrightarrow{\pi} M$ be a vector bundle, and let ∇ be a connection on E . Then we define the *covariant splitting* h_∇ to be the splitting

$$0 \longrightarrow V \longrightarrow TE \xrightarrow{\quad h_\nabla \quad} \pi^* TM \longrightarrow 0$$

defined by

$$h_\nabla(x, X) := (\sigma_*)_{\pi(x)} X,$$

where $\sigma \in \Gamma(E)$ is some section of E satisfying $\sigma(\pi(x)) = x$ and $\nabla \sigma = 0$ on a neighborhood of $\pi(x)$.

To see that h_∇ is indeed a splitting, observe that

$$(\pi_*)_x(\sigma_*)_{\pi(x)}(X) = ((\pi \circ \sigma)_*)_{\pi(x)}X = ((\text{Id}_M)_*)_{\pi(x)}X = X.$$

We denote the image of h_∇ by H , and we call H the *horizontal subbundle* of TE determined by ∇ . As with short exact sequences of vector spaces, the right-splitting h_∇ induces a left-splitting v_∇ as follows. We have $TE = V \oplus H$, and the map $v_\nabla : TE \rightarrow V$ given by projection onto the first factor is called the *vertical projection* determined by ∇ .

Remark 3.2. Proposition 3.4 says that we can use v_∇ to recover ∇ , so connections, vertical projections, and covariant splittings are really just three different ways of looking at the same bit of information.

Let $p \in M$. Since $\pi^{-1}(p)$ is a vector space, for every $x \in \pi^{-1}(p)$ there is a canonical identification $T_x\pi^{-1}(p) \cong \pi^{-1}(p)$. For $y \in \pi^{-1}(p)$, let y_x denote the corresponding vector in $T_x\pi^{-1}(p)$. This notation allows us to more clearly state the following lemma, whose proof is straightforward.

Lemma 3.3. *The map*

$$\begin{aligned} \pi^*E &\rightarrow V \\ (x, y) &\mapsto (x, y_x) \end{aligned}$$

is an isomorphism of vector bundles over E .

By Lemma 3.3, for any section $\sigma : M \rightarrow E$, we have

$$\sigma^*V \cong \sigma^*\pi^*E \cong (\pi \circ \sigma)^*E \cong (\text{Id}_M)^*E = E.$$

Given a section $\sigma : M \rightarrow E$, we abuse notation and let $\sigma_* : TM \rightarrow \sigma^*TE$ denote the bundle homomorphism

$$\sigma_*(p, X) := (p, (\sigma_*)_pX), \quad \text{for } p \in M, X \in T_pM.$$

Proposition 3.4. *Let $E \xrightarrow{\pi} M$ be a vector bundle and let ∇ be a connection on E . Consider the composition*

$$TM \xrightarrow{\sigma_*} \sigma^*TE \xrightarrow{v_\nabla} \sigma^*V \cong E.$$

For any $p \in M$ and $X \in T_pM$, we have $(v_\nabla \circ \sigma_)(p, X) = (\nabla\sigma)_p(X)$.*

Proof. Throughout, we will not write the first coordinate of any $(p, \cdot) \in \sigma^*TE$. Since $TE = V \oplus H$, we also have $\sigma^*TE = \sigma^*V \oplus \sigma^*H$. Therefore we can write $(\sigma_*)_pX = v_\nabla((\sigma_*)_pX) + h_\nabla((\pi_*)_{\sigma(p)}(\sigma_*)_pX)$. Since $\pi_*\sigma_* = (\text{Id}_M)_*$, we have $h_\nabla((\pi_*)_{\sigma(p)}(\sigma_*)_pX) = h_\nabla(X)$. Then $v_\nabla((\sigma_*)_pX) = (\sigma_*)_pX - h_\nabla(X)$. It therefore suffices to show that $(\sigma_*)_pX - h_\nabla(X) = (\nabla\sigma)_p(X)$.

Note that we are writing h_∇ for the map $\sigma^*\pi^*TM = TM \rightarrow \sigma^*TE$, induced by the map with the same name $h_\nabla : \pi^*TM \rightarrow TE$. By Definition 3.1, $h_\nabla(X) = (\tau_*)_pX$, where $\tau(p) = \sigma(p)$ and $\nabla\tau = 0$ in a neighborhood of p . Since $\tau(p) = \sigma(p)$, we have $\sigma - \tau = f\eta$ for some $f \in C^\infty(M)$ with $f(p) = 0$. Then we have

$$\begin{aligned} ((\sigma - \tau)_*)_pX &= ((f\eta)_*)_p(X) \\ &= d_p f(X)\eta(p) + f(p)(\eta_*)_p(X) \\ &= d_p f(X)\eta(p), \end{aligned}$$

and, since $\nabla\tau = 0$ in a neighborhood of p , we have

$$\begin{aligned} (\nabla\sigma)_p(X) &= (\nabla\sigma)_p(X) - (\nabla\tau)_p(X) \\ &= (\nabla(\sigma - \tau))_p(X) \\ &= (\nabla(f\eta))_p(X) \\ &= d_p f(X)\eta(p) + f(p)(\nabla\eta)_p(X) \\ &= d_p f(X)\eta(p). \end{aligned}$$

Therefore $(\nabla\sigma)_p(X) = ((\sigma - \tau)_*)_pX = (\sigma_*)_pX - h_\nabla(X)$, as desired. $\square$

3.2 Structure Groups

In this subsection we introduce the notion of structure group for a vector bundle.

Definition 3.5. Given a topological group G , a *principal G -bundle* over a manifold M is a fiber bundle $P \rightarrow M$ with a fiber-preserving right G -action $P \times G \rightarrow P$ that is free and transitive on each fiber. That is to say, the fibers of P are G -torsors.

I will try to be consistent in making the stylistic choice of writing horizontal lines over things having to do with principal bundles, e.g. the projection $\bar{\pi}$. This is purely stylistic and has no mathematical content. The group G is sometimes said to be the *structure group* of the bundle P , as well as the *structure group* for any associated bundle, defined as follows.

Definition 3.6. Given a topological group G , a principal G -bundle $P \xrightarrow{\bar{\pi}} M$, a space X , and a homomorphism $\rho : G \rightarrow \text{Homeo}(X)$, the *associated bundle* $P \times_{\rho} X \xrightarrow{\pi} M$ is defined by

$$P \times_{\rho} X := (P \times X) / \sim,$$

where $\sim$ is given by $(p, x) \sim (pg^{-1}, \rho(g)x)$ for every $g \in G$. We denote the equivalence class of (p, x) under $\sim$ by $[p, x]_{\rho}$.

One important class of examples of associated bundles is the case where X is a finite-dimensional vector space W , and $\rho : G \rightarrow \text{GL}(W)$.

Example 3.7. Given a rank n vector bundle $E \xrightarrow{\pi} M$, the *frame bundle* $\text{GL}(E)$ is the principal $\text{GL}(n, \mathbb{R})$ -bundle $\text{GL}(E) \rightarrow M$ defined by

$$\text{GL}(E) := \{(v_1, \dots, v_n) \in E^n \mid v_1, \dots, v_n \text{ are in the same fiber of } E \text{ and are a basis for that fiber}\},$$

where we have $(v_1, \dots, v_n)A = (A^{-1}v_1, \dots, A^{-1}v_n)$ for every $A \in \text{GL}(n, \mathbb{R})$, $(v_1, \dots, v_n) \in \text{GL}(E)$. Then we have a bundle isomorphism

$$\Phi : E \rightarrow \text{GL}(E) \times_{\text{Id}_{\text{GL}(n, \mathbb{R})}} \mathbb{R}^n$$

defined as follows. Let $U \subseteq M$ be a chart over which E admits a trivialization $\varphi_U : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$, and let $e_1, \dots, e_n$ be the standard basis for $\mathbb{R}^n$. Then for $x \in \pi^{-1}(U)$, we define

$$\Phi(x) := [(\varphi_U^{-1}(\pi(x), e_1), \dots, \varphi_U^{-1}(\pi(x), e_n)), \text{proj}_{\mathbb{R}^n}(\varphi_U(x)))]_{\text{Id}_{\text{GL}(n, \mathbb{R})}},$$

where $\text{proj}_{\mathbb{R}^n}$ denotes the projection onto the $\mathbb{R}^n$ factor. One can check that $\sim$ is indeed exactly the equivalence relation we want in order for this map to be a well-defined isomorphism.

Example 3.8. Given a Lie group G with Lie algebra $\mathfrak{g}$ and a principal G -bundle $P \rightarrow M$, the *adjoint bundle* is the associated vector bundle

$$P \times_{\text{Ad}_G} \mathfrak{g},$$

where $\text{Ad}_G : G \rightarrow \text{GL}(\mathfrak{g})$ is the adjoint representation $g : x \mapsto (R_{g^{-1}})_*(L_g)_*x$ for $g \in G$, $x \in \mathfrak{g}$, and $R_{g^{-1}} : h \mapsto hg^{-1}$ and $L_g : h \mapsto gh$ denote the left- and right-multiplication maps $G \rightarrow G$.

Example 3.9. Let M be a connected space and let $p \in M$. Observe that the universal cover $\widetilde{M}$ of M is a principal $\pi_1(M, p)$ -bundle via the right-action of $\pi_1(M, p)$ on $\widetilde{M}$ by deck transformations. Given a homomorphism $\rho : \pi_1(M, p) \rightarrow G$ for a topological group G , we can form the associated bundle

$$P_{\rho} := \widetilde{M} \times_{\rho} G.$$

Note that P_{ρ} is itself a principal G -bundle via the right-action $[x, g]_{\rho}h = [x, gh]_{\rho}$ for all $h \in G$.

We will define $H_{\rho}^1(\mathfrak{g})$ to be the twisted cohomology of a particular flat connection on $P_{\rho} \times_{\text{Ad}_G} \mathfrak{g}$. This connection will be induced by a connection on P_{ρ} , so we must first develop the notion of a connection on a principal bundle.

3.3 Connections on Principal Bundles

From now on, we will consider Lie groups rather than general topological groups. If G is a Lie group, then we require that the G -action on a principal G bundle be an action by diffeomorphisms. We will define connections on principal bundles after the manner of Definition 3.1. By Remark 3.2, this approach will indeed give us something that should be called a “connection.” Given a Lie group G and a principal G -bundle $P \xrightarrow{\pi} M$, define the *vertical subbundle* $\bar{V}$ of TP in the same way as in the beginning of section 3.1, $\bar{V} := \bigcup_{x \in P} \ker(\pi_*)_x$.

Definition 3.10. Given a Lie group G and a principal G -bundle $P \rightarrow M$, a *connection* $\bar{h}$ on P is a splitting

$$0 \longrightarrow \bar{V} \longrightarrow TP \xrightarrow{\quad \bar{h} \quad} \pi^*TM \longrightarrow 0$$

that is G -equivariant in the sense that $\bar{h}(pg, X) = ((R_g)_*)_p \bar{h}(p, X)$ for every $g \in G$, where $R_g : P \rightarrow P$ is the right-multiplication map $p \mapsto pg$.

We denote the image of $\bar{h}$ by $\bar{H}$, and we call $\bar{H}$ the *horizontal subbundle* of TE .

Proposition 3.11. *Given a Lie group G , a principal G -bundle $P \rightarrow M$, a finite-dimensional vector space W , and an injective homomorphism $\rho : G \rightarrow \mathrm{GL}(W)$, let $E = P \times_\rho W$. Then a connection $\bar{h}$ on P induces a splitting h of the sequence*

$$0 \longrightarrow V \longrightarrow TE \xrightarrow{\quad h \quad} \pi^*TM \longrightarrow 0$$

defined by

$$h([p, w]_\rho, X) = (f_w)_* \bar{h}(p, X),$$

where $f_w : P \rightarrow E$ is the map $p \mapsto [p, w]_\rho$.

Proof. We first show that h is well-defined. Given $(p, w) \in P \times W$ and $g \in G$, consider $(pg^{-1}, \rho(g)w)$. We want to show that $(f_w)_* \bar{h}(p, X) = (f_{\rho(g)w})_* \bar{h}(pg^{-1}, X)$. Given $q \in P$, we have $f_{\rho(g)w}(q) = [q, \rho(g)w]_\rho = [qg, w]_\rho$, and hence $f_{\rho(g)w} = f_w \circ R_g$, where $R_g : P \rightarrow P$ is the right-multiplication map $q \mapsto qg$. Therefore

$$\begin{aligned} (f_{\rho(g)w})_* \bar{h}(pg^{-1}, X) &= (f_w)_* (R_g)_* \bar{h}(pg^{-1}, X) \\ &= (f_w)_* \bar{h}(pg^{-1}g, X) \quad \text{by the } G\text{-equivariance of } \bar{h} \\ &= (f_w)_* \bar{h}(p, X). \end{aligned}$$

Therefore h is well-defined.

To show that h is a splitting, we must show that $\pi_* \circ h = \mathrm{Id}_{\pi^*TM}$. Note that f_w is fiber-preserving for any $w \in W$, and so $\pi \circ f_w = \pi$. Since we are considering π_* as a map $TE \rightarrow \pi^*TM$ and π_* as a map $TP \rightarrow \pi^*TM$, the equation $\pi \circ f_w = \pi$ gives us a commutative diagram

$$\begin{array}{ccc} TP & \xrightarrow{(f_w)_*} & TE \\ \pi_* \downarrow & & \downarrow \pi_* \\ \pi^*TM & \xrightarrow{g: (p, X) \mapsto ([p, w]_\rho, X)} & \pi^*TM \end{array}$$

where the diagram chase looks like

$$\begin{array}{ccc} (p, Y) & \xrightarrow{\quad} & (p, (f_w)_* Y) \\ \downarrow & & \downarrow \\ ([p, w]_\rho, \pi_* Y) & \xrightarrow{\quad} & ([p, w]_\rho, (\pi \circ f_w)_* Y) = ([p, w]_\rho, \pi_* Y). \end{array}$$

Therefore for any $[p, w]_\rho \in E$, $X \in T_{\pi([p, w]_\rho)}M$, we have

$$\begin{aligned} (\pi_* \circ h)([p, w]_\rho, X) &= (\pi_* \circ (f_w)_* \circ \bar{h})(p, X) \\ &= (g \circ \pi_* \circ \bar{h})(p, X) \\ &= g(p, X) \\ &= ([p, w]_\rho, X). \end{aligned}$$

Therefore h is a splitting. $\square$

Definition 3.12. Suppose a vector bundle E can be written as $E = P \times_\rho W$ for some principal G -bundle P , some finite-dimensional vector space W , some injective homomorphism $\rho : G \rightarrow \text{GL}(W)$. Then a connection ∇ on E is said to be a G -connection if the covariant splitting h_∇ is induced by a connection $\bar{h}$ on P in the manner of Proposition 3.11.

3.4 Holonomy for Principal Bundles

Just as in the case of vector bundles, a connection $\bar{h}$ on a principal bundle $P \rightarrow M$ determines a way of uniquely lifting paths from M to P given an initial lift. We say that a path $\eta : [0, 1] \rightarrow P$ is *horizontal* if $\eta'(t) \in \bar{H}$ for every $t \in [0, 1]$.

Proposition 3.13. Let G be a Lie group and $P \xrightarrow{\pi} M$. Given a path $\gamma : [0, 1] \rightarrow M$ and some $p \in \pi^{-1}(\gamma(0))$, there is a unique horizontal lift $\tilde{\gamma} : [0, 1] \rightarrow P$ satisfying $\tilde{\gamma}(0) = p$.

Proposition 3.13 is proved in the same way as Proposition 2.9: by invoking the existence and uniqueness theorem for ODEs. We can also make a definition for principal bundles similar in spirit to Definition 2.10. Given a loop $\gamma : [0, 1] \rightarrow M$, let $\text{Hol}^{\bar{h}}(\gamma)(p) := \tilde{\gamma}(1)$, where $\tilde{\gamma}$ is the unique horizontal lift of γ satisfying $\tilde{\gamma}(0) = p$.

Definition 3.14. Let G be a Lie group, $P \xrightarrow{\pi} M$ a principal G -bundle, and $\bar{h}$ a connection on P . Given a point $p \in P$, the *holonomy group* $\text{Hol}_p^{\bar{h}}$ is the subgroup of G given by

$$\text{Hol}_p^{\bar{h}} := \{g \in G \mid \exists \text{ a loop } \gamma : [0, 1] \rightarrow M \text{ with } \gamma(0) = \gamma(1) = \pi(p) \text{ s.t. } \text{Hol}^{\bar{h}}(\gamma)(p) = pg\}.$$

Theorem 3.15. Recall the principal bundle P_ρ from Example 3.9, and suppose that M a smooth manifold and G is a Lie group. Then there exists a unique connection $\bar{h}_\rho$ on P_ρ such that $\text{Hol}_p^{\bar{h}_\rho} = \rho(\pi_1(M, p))$.

Proof. Let $\tau : \widetilde{M} \times G \rightarrow \widetilde{M} \times_\rho G = P_\rho$ be the quotient projection, let $\pi : \widetilde{M} \times_\rho G \rightarrow M$ be the bundle projection, and let $\varphi : \widetilde{M} \rightarrow M$ be the covering map. Since φ is a local diffeomorphism, for any $x \in \widetilde{M}$, $X \in T_{\varphi(x)}M$ there is a unique $\tilde{X}_x \in T_x\widetilde{M}$ with $\varphi_*\tilde{X}_x = X$. Define $\bar{k} : (\pi \circ \tau)^*TM \rightarrow T(\widetilde{M} \times G)$ by $\bar{k}((x, g), X) = ((x, g), (\tilde{X}_x, 0))$. We claim that there is a unique factorization

$$\begin{array}{ccc} (\pi \circ \tau)^*TM & \xrightarrow{\tau_* \circ \bar{k}} & T(\widetilde{M} \times_\rho G) \\ & \searrow \bar{\tau} & \nearrow \exists! \bar{h}_\rho \\ & \pi^*TM & \end{array}$$

where $\bar{\tau} : (\pi \circ \tau)^*TM \rightarrow \pi^*TM$ is the canonical map $((x, g), X) \mapsto ([x, g]_\rho, X)$ induced by the pullback τ^* .

Given $((x, g), X) \in (\pi \circ \tau)^*TM$, we have $(\tau_* \circ \bar{k})((x, g), X) = ([x, g]_\rho, \tau_*(\tilde{X}_x, 0))$. To show that $\tau_* \circ \bar{k}$ factors through $\bar{\tau}$, we must show that $(\tau_* \circ \bar{k})((x, g), X) = (\tau_* \circ \bar{k})((p\gamma^{-1}, \rho(\gamma)g), X)$ for every $\gamma \in \pi_1(M, p)$. Since the action of $\pi_1(M, p)$ is by deck transformations, we have $\tilde{X}_x = (R_{\gamma^{-1}})_*\tilde{X}_{x\gamma^{-1}}$ for any $\gamma \in \pi_1(M, p)$,

where $R_{\gamma^{-1}} : \widetilde{M} \rightarrow \widetilde{M}$ is right-multiplication $x \mapsto x\gamma^{-1}$. Furthermore, we have $\tau \circ (R_{\gamma^{-1}} \times L_{\rho(\gamma)}) = \tau$ by definition, where $L_{\rho(\gamma)} : G \rightarrow G$ is left-multiplication $g \mapsto \rho(\gamma)g$. Therefore

$$\begin{aligned}
(\tau_* \circ \bar{k})((x, g), X) &= ([x, g]_\rho, \tau_*(\widetilde{X}_x, 0)) \\
&= ([x\gamma^{-1}, \rho(\gamma)g]_\rho, \tau_*(R_{\gamma^{-1}} \times L_{\rho(\gamma)})_*(X_x, 0)) \\
&= ([x\gamma^{-1}, \rho(\gamma)g]_\rho, \tau_*((R_{\gamma^{-1}})_*X_x, (L_{\rho(\gamma)})_*0)) \\
&= ([x\gamma^{-1}, \rho(\gamma)g]_\rho, \tau_*(X_{x\gamma^{-1}}, 0)) \\
&= (\tau_* \circ \bar{k})((x\gamma^{-1}, \rho(\gamma)g), X),
\end{aligned}$$

as desired. Hence $\tau_* \circ \bar{k}$ factors as $\bar{h}_\rho \circ \bar{\tau}$ for some $\bar{h}_\rho$. Furthermore, since $\bar{\tau}$ is an epimorphism, the map $\bar{h}_\rho$ is unique.

We now claim that $\bar{h}_\rho$ is a G -equivariant splitting

$$0 \longrightarrow \bar{V} \longrightarrow TP_\rho \xrightarrow[\pi_*]{\bar{h}_\rho} \pi^*TM \longrightarrow 0.$$

Observe that $(\pi_* \circ \tau_*)((x, g), (Y, A)) = ([x, g]_\rho, \varphi_*Y)$ for every $((x, g), (Y, A)) \in T(\widetilde{M} \times G)$, where we consider π_* as the map $TP_\rho \xrightarrow{\pi_*} \pi^*TM$. Therefore, for every $([x, g]_\rho, X) \in \pi^*TM$, we have

$$\begin{aligned}
(\pi_* \circ \bar{h}_\rho)([x, g]_\rho, X) &= (\pi_* \circ \bar{h}_\rho \circ \bar{\tau})((x, g), X) \\
&= (\pi_* \circ \tau_* \circ \bar{k})((x, g), X) \\
&= (\pi_* \circ \tau_*)((x, g), (\widetilde{X}_x, 0)) \\
&= ([x, g]_\rho, \varphi_*\widetilde{X}_x) \\
&= ([x, g]_\rho, X).
\end{aligned}$$

Therefore $\bar{h}_\rho$ is a splitting, as claimed. To see that $\bar{h}_\rho$ is G -equivariant, let $h \in G$. Then

$$\begin{aligned}
\bar{h}_\rho([x, gh]_\rho, X) &= (\tau_* \circ \bar{k})((x, gh), X) \\
&= \tau_*((x, gh), (\widetilde{X}_x, 0)) \\
&= \tau_*(R_h(x, g), (R_h)_*(\widetilde{X}_x, 0)) \\
&= (\tau_* \circ (R_h)_*)((x, g), (\widetilde{X}_x, 0)) \\
&= ((R_h)_* \circ \tau_*)((x, g), (\widetilde{X}_x, 0)) \quad \text{by the } G\text{-equivariance of } \tau \\
&= (R_h)_*\bar{h}([x, g]_\rho, X).
\end{aligned}$$

Therefore $\bar{h}_\rho$ is a G -equivariant splitting of $0 \rightarrow V \rightarrow TP_\rho \rightarrow \pi^*TM \rightarrow 0$, and hence $\bar{h}_\rho$ is a connection on P_ρ .

We now want to show that $\text{Hol}_p^{\bar{h}_\rho} = \rho(\pi_1(M, p))$. Let $\gamma \in \pi_1(M, p)$, and let $\tilde{\gamma} : [0, 1] \rightarrow \widetilde{M}$ be the lift of some representative of γ satisfying $\varphi(\tilde{\gamma}(0)) = p$. The path $\tilde{\gamma}$ determines a path $\eta : [0, 1] \rightarrow \widetilde{M} \times G$ given by $\eta(t) = (\tilde{\gamma}(t), 1)$, which descends to a path $\tilde{\eta} : [0, 1] \rightarrow \widetilde{M} \times_\rho G = P_\rho$ given by $\tilde{\eta}(t) = \tau(\eta(t)) = [\tilde{\gamma}(t), 1]_\rho$. Notice that $\tilde{\eta}$ is a horizontal lift of γ to P_ρ , since $\eta'(t) = (\tilde{\gamma}'(t), 0)$ for every $t \in [0, 1]$, and hence $\tilde{\eta}'(t)$ lies in the image of $\bar{h}_\rho$ (i.e. the horizontal subbundle of TP_ρ). By the definition of the $\pi_1(M, p)$ -action on $\widetilde{M}$, we have $\tilde{\gamma}(1) = \tilde{\gamma}(0)\gamma$ (this notation is a bit confusing here; this is the right-action of the group element

$\gamma \in \pi_1(M, p)$ on the point $\tilde{\gamma}(0) \in \widetilde{M}$. Therefore we have

$$\begin{aligned}\eta(1) &= [\tilde{\gamma}(1), 1]_\rho \\ &= [\tilde{\gamma}(0)\gamma, 1]_\rho \\ &= [\tilde{\gamma}(0), \rho(\gamma)]_\rho \\ &= [\tilde{\gamma}(0), 1]_\rho \rho(\gamma) \\ &= \eta(0)\rho(\gamma),\end{aligned}$$

and so $\rho(\gamma) \in \text{Hol}_p^{\bar{h}_\rho}$. Therefore $\rho(\pi_1(M, p)) \subseteq \text{Hol}_p^{\bar{h}_\rho}$. Furthermore, since the above construction depended only on the homotopy class of a loop in M at p , we have in fact enumerated all elements of $\text{Hol}_p^{\bar{h}_\rho}$, and so $\text{Hol}_p^{\bar{h}_\rho} = \rho(\pi_1(M, p))$.

We omit the proof that $\bar{h}_\rho$ is in fact the *unique* connection on P_ρ such that $\text{Hol}_p^{\bar{h}_\rho} = \rho(\pi_1(M, p))$. $\square$

3.5 Tying it All Together

The last bit of heavy lifting we need to relate a connection on principal bundle to the twisted cohomology of an associated vector bundle is the Ambrose-Singer Theorem. The following proposition can be considered to be a corollary of the Ambrose-Singer Theorem, and we do not prove this here. See §II.8 of [2] and §9.5 of [5] for treatments of the Ambrose-Singer Theorem.

Proposition 3.16. *If $\text{Hol}_p^{\bar{h}}$ is a discrete group, then the G -connection induced by $\bar{h}$ is flat.*

This is a corollary in the following sense. There is a corresponding notion of curvature for connections on principal bundles, and furthermore the curvature and holonomy of a G -connection ∇ are determined by the principal connection inducing ∇ . The Ambrose-Singer Theorem implies that a principal connection with discrete holonomy group has vanishing curvature, and hence ∇ also has vanishing curvature.

Together, Theorem 3.15 and Proposition 3.16 give the following theorem.

Theorem 3.17. *Let P_ρ be the principal bundle from Example 3.9, and let ∇^ρ be the G -connection on $P_\rho \times_{\text{Ad}_G} \mathfrak{g}$ induced by $\bar{h}_\rho$. Then ∇^ρ is flat.*

Theorem 3.17 implies that we can make the following definition.

Definition 3.18. Let S be a surface with $p \in S$, let G be a linear Lie group with Lie algebra $\mathfrak{g}$, and let $\rho : \pi_1(S, p) \rightarrow G$ be a representation. We define

$$H_\rho^n(\mathfrak{g}) := H_{\nabla^\rho}^n(S, P_\rho \times_{\text{Ad}_G} \mathfrak{g}).$$

At this point, it is non-obvious why an element of $H_\rho^1(\mathfrak{g})$ should represent an “infinitesimal direction in which ρ may vary.” In Chapter 4 of [3], a nice explanation is given of why an element of $H_\rho^1(\mathfrak{g})$ should represent an infinitesimal direction in which ∇^ρ may vary. The idea is that since the space of connections on a bundle $E \rightarrow M$ is an $\Omega^1(M, \text{End}(E))$ -torsor, a tangent vector in the space of connections at the point ∇^ρ should also be given by an element $\dot{\nabla}$ of $\Omega^1(M, \text{End}(E))$. Furthermore, since we are considering G -connections, we should restrict our attention to $P_\rho \times_{\text{Ad}_G} \mathfrak{g} \subseteq \text{End}(E)$ (cf. Definition 3.3.18 of Labourie; see Proposition A.7 of these notes for the proof of this inclusion). One then shows that if $\dot{\nabla}$ is a variation through flat connections, then $d_{\nabla^\rho} \dot{\nabla} = 0$, i.e. $\dot{\nabla}$ is closed, and if $\dot{\nabla}$ comes from the action of the gauge group (see Definition A.1) by conjugation, then $\dot{\nabla}$ is exact. Given a connection ∇ , the conjugate of ∇ by a member of the gauge group has holonomy group conjugate to the holonomy group of ∇ . Since we quotient by conjugation when forming $\text{Rep}(\pi_1(S), G)$, we want to quotient out by connections that come from the action of the gauge group. That is to say, we want to consider closed $\dot{\nabla}$ only up to exact $\dot{\nabla}$. This is precisely the construction of $H_\rho^1(\mathfrak{g})$.

A Gauge Transformations and Reductions of Structure Groups

Definition A.1. Given a topological group G and a principal G -bundle $P \rightarrow M$, the *gauge group* $\mathcal{G}(P)$ is the group of bundle automorphisms $\varphi : P \rightarrow P$ that are G -equivariant, i.e. $\varphi(pg) = \varphi(p)g^{-1}$ for every $p \in P, g \in G$. The elements of $\mathcal{G}(P)$ are called *gauge transformations* of P .

Definition A.2. Given topological groups H, G , a principal H -bundle $Q \rightarrow M$, a principal G -bundle $P \rightarrow M$, and an injective homomorphism $\iota : H \rightarrow G$, a *reduction of the structure group of P from G to H given by Q* and ι is a fiber-preserving embedding $\varphi : Q \hookrightarrow P$ that is H -equivariant, i.e. $\varphi(ph) = \varphi(p)\iota(h)$ for every $p \in P, h \in H$.

Remark A.3. It is interesting to notice that a gauge transformation is the same thing as a reduction of the structure group of P from G to G given by P and the identity homomorphism $\text{Id}_G : G \rightarrow G$.

Proposition A.4. Given topological groups H, G , a principal H -bundle $Q \rightarrow M$, a principal G -bundle $P \rightarrow M$, an injective homomorphism $\iota : H \rightarrow G$, a finite-dimensional vector space W , and an injective homomorphism $\rho : G \rightarrow \text{GL}(W)$, a reduction $\varphi : Q \hookrightarrow P$ given by Q and ι induces an isomorphism $\Phi : Q \times_{\rho \circ \iota} W \rightarrow P \times_{\rho} W$ given by

$$\Phi([q, w]_{\rho \circ \iota}) := [\varphi(q), w]_{\rho}.$$

Proof. We first show that Φ is well-defined. If we write $[q, w]_{\rho \circ \iota}$ instead as $[qh^{-1}, \rho(\iota(h))w]_{\rho \circ \iota}$, we have

$$\begin{aligned} \Phi([qh^{-1}, \rho(\iota(h))w]_{\rho \circ \iota}) &= [\varphi(qh^{-1}), \rho(\iota(h))w]_{\rho} \\ &= [\varphi(q)h^{-1}, \rho(\iota(h))w]_{\rho} \\ &= [\varphi(q), w]_{\rho}, \end{aligned}$$

and so Φ is well-defined.

We now show that Φ is injective. If $[\varphi(q), w]_{\rho} = [\varphi(q'), w']_{\rho}$, then there is some $g \in G$ with $(\varphi(q)g^{-1}, \rho(g)w) = (\varphi(q'), w')$. Since H acts transitively on the fibers of Q , there is some $h \in H$ with $q' = qh^{-1}$. Hence $\varphi(q') = \varphi(qh^{-1}) = \varphi(q)g^{-1}$. Since G acts freely on the fibers of P , it follows that $\iota(h) = g$, and hence $[q, w]_{\rho \circ \iota} = [q', w']_{\rho \circ \iota}$. Thus Φ is injective.

We now show that Φ is surjective. Let $[p, w]_{\rho} \in P \times_{\rho} W$, and let $q \in Q$. Since G acts transitively on the fibers of P , there is some $g \in G$ with $(pg^{-1}, \rho(g)w) = (\varphi(q), \rho(g)w)$. Hence $[p, w]_{\rho} = \Phi([q, \rho(g)w]_{\rho \circ \iota})$, and so Φ is surjective. We have checked that Φ is a well-defined bijection, and so it is now clear that Φ is in fact a vector bundle isomorphism. $\square$

Corollary A.5. Given a topological group G , a principal G -bundle $P \rightarrow M$, a finite-dimensional vector space W , and an injective homomorphism $\rho : G \rightarrow \text{GL}(W)$, the gauge group $\mathcal{G}(P)$ of P acts by vector bundle automorphisms on $P \times_{\rho} W$.

Proof. This is an immediate consequence of Remark A.3 and Proposition A.4. $\square$

Lemma A.6. Given a rank n vector bundle $E \rightarrow M$, we have a bundle isomorphism

$$\text{End}(E) \cong \text{GL}(E) \times_{\text{Ad}_{\text{GL}(n, \mathbb{R})}} \text{End}(\mathbb{R}^n),$$

where $\text{Ad}_{\text{GL}(n, \mathbb{R})} : \text{GL}(n, \mathbb{R}) \rightarrow \text{GL}(\text{End}(\mathbb{R}^n))$ denotes the representation $T : A \mapsto TAT^{-1}$ for $T \in \text{GL}(n, \mathbb{R})$, $A \in \text{End}(\mathbb{R}^n)$.

Proof. Recall that any rank k vector bundle $F \xrightarrow{\pi} M$ is uniquely determined by its change-of-trivialization maps (i.e. cocycles) $\varphi_{UV} : U \cap V \rightarrow \text{GL}(\mathbb{R}^k)$, where $U, V \subseteq M$ are charts over which the F admits trivializations $\varphi_U : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$, $\varphi_V : \pi^{-1}(V) \rightarrow V \times \mathbb{R}^k$, and $\varphi_{UV} = \text{proj}_{\mathbb{R}^k} \circ \varphi_V \circ \varphi_U^{-1}$, where $\text{proj}_{\mathbb{R}^k}$ denotes the projection onto the $\mathbb{R}^k$ factor. It therefore suffices to show that $\text{End}(E)$ and $\text{GL}(E) \times_{\text{Ad}_{\text{GL}(n, \mathbb{R})}} \text{End}(\mathbb{R}^n)$ are determined by the same change-of-trivialization maps.

Let $U, V \subseteq M$ be neighborhoods over which E is trivial, and let $\varphi_{UV} : U \cap V \rightarrow \mathrm{GL}(\mathbb{R}^n)$ be a change-of-trivialization for E . Then $\mathrm{GL}(E) \times_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n)$ has by definition¹ a change-of-trivialization ψ_{UV} given by

$$\begin{aligned} \psi_{UV} : [(v_1, \dots, v_n), A]_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} &\mapsto [(\varphi_{UV}(p)^{-1}v_1, \dots, \varphi_{UV}(p)^{-1}v_n), A]_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \\ &= [(v_1, \dots, v_n)\varphi_{UV}(p), A]_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \\ &= [(v_1, \dots, v_n), \varphi_{UV}(p)A\varphi_{UV}(p)^{-1}]_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \end{aligned}$$

for any point $[(v_1, \dots, v_n), A]_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \in \mathrm{GL}(E) \times_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n)$ lying in the fiber over $p \in U \cap V$. Thus we have

$$\begin{aligned} \psi_{UV} : U \cap V &\rightarrow \mathrm{GL}(\mathrm{End}(\mathbb{R}^n)) \\ p &\mapsto \mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}(\varphi_{UV}(p)), \end{aligned}$$

for every $U, V \subseteq M$ over which E is trivialized, and these ψ_{UV} are precisely the changes-of-trivialization used to define $\mathrm{End}(E)$. Therefore we have

$$\mathrm{End}(E) \cong \mathrm{GL}(E) \times_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n),$$

as desired. □

Proposition A.7. *Given a Lie group G with Lie algebra $\mathfrak{g}$, a principal G -bundle $P \rightarrow M$, a rank n vector bundle $E \rightarrow M$, an injective homomorphism $\rho : G \rightarrow \mathrm{GL}(n, \mathbb{R})$, and a reduction of structure group $\varphi : P \rightarrow \mathrm{GL}(E)$ determined by ρ , there is an inclusion of bundles*

$$P \times_{\mathrm{Ad}_G} \mathfrak{g} \hookrightarrow \mathrm{End}(E)$$

Proof. By Lemma A.6, we have

$$\mathrm{End}(E) \cong \mathrm{GL}(E) \times_{\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n),$$

and by Proposition A.4, we have an isomorphism

$$\mathrm{GL}(E) \times_{\mathrm{Ad}} \mathrm{End}(\mathbb{R}^n) \cong P \times_{\rho \circ \mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n).$$

Then the derivative of ρ is an injective homomorphism $\rho_* : \mathfrak{g} \hookrightarrow \mathfrak{gl}(n) = \mathrm{End}(\mathbb{R}^n)$, and hence induces a bundle inclusion

$$P \times_{\mathrm{Ad}_G} \mathfrak{g} \hookrightarrow P \times_{\rho \circ \mathrm{Ad}_{\mathrm{GL}(n, \mathbb{R})}} \mathrm{End}(\mathbb{R}^n) \cong \mathrm{End}(E),$$

as desired. □

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¹This can be seen by making a slight modification to Example 3.7.